3-) a) v is a unit vector in R^n

H is a $n \times n$ matrix

therefore, $H = I - 2vv^T$, As v is a unit vector

moreover, $\|v\| = 1$

a-) $H^2 = H \cdot H = (I - 2vv^T)(I - 2vv^T)$

$= I^2 - I - 2vv^T - 2vv^T \cdot I + 4vv^T \cdot vv^T$

$= I(I - 2vv^T - 2vv^T) + 4v \cdot \|v\|^2 \cdot v^T$

$= I - 2vv^T - 2vv^T + 4v \cdot \|v\|^2 \cdot v^T$

so $\|v\|^2 = v^T \cdot v$

$\|v\| = \sqrt{v^T \cdot v}$ $\|v\| = 1$

consequently $= H^2 = I - 4vv^T + 4vv^T$

$H^2 = I - 4(1) + 4(1)$

$\underline{H^2 = I}$

b-) $MM^T = (I - 2vv^T)(I - 2vv^T)^T$

$= (I - 2vv^T) \cdot (I^T - (2vv^T)^T)$

$= I - 2vv^T) \cdot (I - 2((v^T)^T v^T)$

$= (I - 2vv^T)(I - 2vv^T)$

$= I^2 - 2vv^T - 2vv^T + 4vv^T \cdot vv^T$

$= I - 4vv^T + 4v\|v\|^2 \cdot v^T$

$= I - 4vv^T + 4vv^T$

$= I - 4(1) \cdot (4(1)$

$MM^T = I$

$[A - B]^T = A^T - B^T$

$(AB)^T = B^T \cdot A^T$

$((v)^T)^T = v$

$\|v\| = 1$

consequently;

$M^T M = I$

Therefore $= MM^T = M^T M = I$

M is ortho. matrix

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c-) As a matrix, the vector(non-zero) must be eigen-vector (suppose v) of A equal to eigenvalue λ if $\underline{Av = \lambda v}$

Firstly: $Hv = (I - 2uu^T) \cdot v$

$= Iv - 2uu^T v$

$= v - 2u\|u\|^2$ because of I is identity matrix

I is identity matrix, therefore .I has eigenvalue of I and $Iv = 1 \cdot v = v$

and $\|u\| = \sqrt{u^T \cdot u}$.

u is unit vector, thus, $\|u\| = 1 = \sqrt{u^T u}$, moreover

$Hv = v - 2v \cdot 1 = -v$

$Hv = -v = (-1) \cdot v$.

according to this equation, we have $\Rightarrow Hv = \lambda v$, where $\lambda = -1$

consequently is an eigenvector of H and equal to eigenvalue $v\bar{v} = \lambda = -1$

d-) $H = (I - 2uu^T)$

$H^T = (I - 2uu^T)^T$

$= I^T - 2(uu^T)^T$

$= I - 2(u^T)^T \cdot u^T$

$= I - 2uu^T = H$

So, H is a symetric matrix and There is a theorem which related that

(where Q is an orthagonal matrix

$H = C \Lambda C^T$

$= C \Lambda C^{-1}$

$= (C^{-1})^{-1} \Lambda C^{-1}$

$= C^{-1} \Lambda C$

$As = CC^T = I = C^T = C^{-1}$

Therefore H is diagonalizable

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On my honor, I have neither given nor received any unauthorized and/or inappropriate assistace for all sessions of this exam. The work done on this exam is totally my own. I understand that by the School code violation of these principles will lead to a zero grade and is subject to harsh discipline issues." 150 719 709 Murat Sort YILDIRAN

1.) consider the matrix $\begin{bmatrix} 1 & \alpha & -4 & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$ Some of the elimination matrices are given below

$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ -3 & 1 & 0 & 0 \end{bmatrix}$, $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

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2-) a-) Becouse at the vector space with vector satisfying $x-5y=0$ is dimension, R^n must be $\underline{\underline{n \geq 5}}$

4-) a-)

$$\begin{bmatrix} w_{k+1} \\ v_{k+1} \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix}$$

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